

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EST130**Course Name: BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING****PART I: BASIC ELECTRICAL ENGINEERING****(2019 Scheme)**

Max. Marks:50

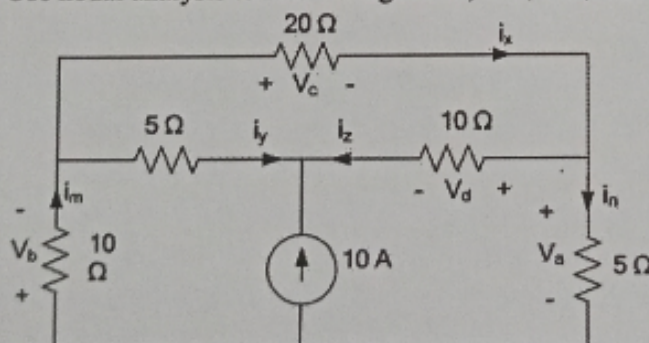
Duration: 90 min

PART A*Answer all questions, each carries 4 marks.*

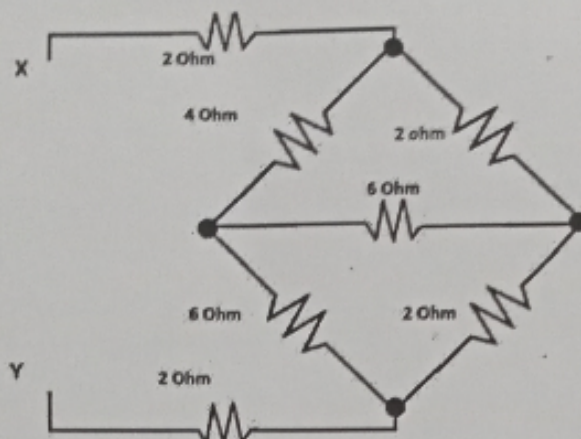
- 1 A coil of 180 turns is linked with a flux of 0.03 Wb when carrying a current of 10A. Calculate the inductance of the coil. If the current is uniformly reversed in 0.04 sec, calculate the emf induced in the coil.
- 2 An alternating current is represented by $i(t) = 14.14 \sin(377t)$. Find (i)rms value (ii) frequency (iii)time period and (iv)instantaneous value of the current at $t=3\text{ms}$.
- 3 Derive an expression for the energy stored in an inductor.
- 4 Derive the expression for the current in an ac series RLC circuit.
- 5 A resistance of 10Ω , an inductance of 0.3 H, and a capacitance of $100\mu\text{F}$ are connected in series across 230V, 50 Hz single phase power supply. Calculate the impedance, current through and power factor of the circuit. (5x4=20)

PART B*Answer one full question from each module, each question carries 10 marks***Module-I**

- 6 Use nodal analysis to find voltages V_a, V_b, V_c, V_d . (10)

**OR**

- 7 Find the equivalent resistance between terminal X-Y in the network (10)



Module-II

- 8 An iron ring has a cross section area of 3 cm^2 and a mean diameter of 25 cm. An air gap of 0.5mm is cut across the section of the ring. The ring is wound with a coil of 200 turns through which a current of 3A is passed. If the total magnetic flux is 0.28 mWb, find the relative permeability of iron, assuming no magnetic leakage. (10)

OR

- 9 a) The instantaneous value of an alternating voltage is given by $v=110 \sin 314t$. Find the angular velocity, frequency, and time period of the voltage. (6)
- b) Differentiate between statically and dynamically induced emfs. (4)

Module-III

- 10 A resistance of 10Ω , an inductance of 0.3 H and a capacitance of $100\mu\text{F}$ are connected in series across 230V, 50 Hz single phase supply. Calculate:
i) the impedance of the circuit ii) Current through the circuit iii) Voltage across R, L and C, and iv) Power consumed by the circuit. (10)

OR

- 11 A balanced delta connected 3 phase load is fed from a 3 phase, 400 V, 50 Hz power supply. The line current is 20A and the total power absorbed by the load is 10kW. Calculate (i) the impedance in each branch (ii) the power factor and (ii) the total power consumed if the same impedances are star connected. (10)

BEE, Answer Key December 2021 ^①

1. $N = 180$

$$\phi = 0.03 \text{ Wb}$$

$$I = 10 \text{ A}$$

$$L = \frac{N\phi}{I} = \frac{180 \times 0.03}{10} = \underline{\underline{0.54 \text{ H}}}$$

$$\begin{aligned} e &= -L \frac{dI}{dt} = 0.54 \times \frac{(-10 - 10)}{0.04} \\ &= -0.54 \times \frac{-20}{0.04} \\ &= \underline{\underline{270 \text{ V}}} \end{aligned}$$

2. $i(t) = 14.14 \sin(377t)$

Comparing with the standard value

$$i(t) = I_m \sin \omega t$$

$$I_m = 14.14 \text{ A}; \quad \omega = 377.$$

$$1) I_{\text{rms}} = \frac{I_m}{\sqrt{2}} = \frac{14.14}{\sqrt{2}} = \underline{\underline{10 \text{ A}}}$$

$$2) \omega = 2\pi f; \quad f = \frac{\omega}{2\pi} = \frac{377}{2 \times 3.14} = 60 \text{ Hz}$$

$$3) T = \frac{1}{f} = \frac{1}{60} = \underline{\underline{0.0167 \text{ s}}}$$

4) i when $t = 3 \text{ ms}$.

$$i = 14.14 \sin(377 \times 3 \times 10^{-3})$$

$$= 14.14 \sin(1.131) \leftarrow \text{here } \theta \text{ in radians}$$

$$= 14.14 \sin\left(1.131 \times \frac{180}{\pi}\right) \leftarrow \theta \text{ in degree}$$

$$= \underline{\underline{12.7978 \text{ A}}}$$

(2)

3. Energy stored in the inductor

In an inductor, if a current is made to pass through it, an electromagnetic field is formed. A change in the magnitude of the current changes the electromagnetic field, which induces a voltage across the coil according to Faraday's Laws of electromagnetic Induction,

$$V = L \frac{di}{dt}$$

V - voltage across inductor in volts,

i - current through inductor in amps.

$$\therefore di = \frac{V}{L} dt$$

Power absorbed by inductor is

$$P = Vi$$

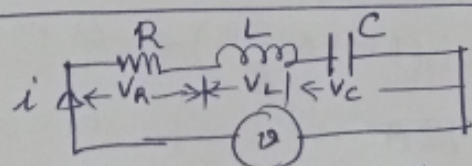
$$= L \frac{di}{dt} \times i = L i \frac{di}{dt} \text{ watts}$$

Energy stored in the inductor

$$W = \int_0^t P \cdot dt = \int_0^t L i \frac{di}{dt} \times dt = \int_0^t L i di$$

$$\boxed{W = \frac{Li^2}{2}} \quad \boxed{\text{energy stored} = \frac{1}{2} LI^2}$$

4. Current in RLC circuit

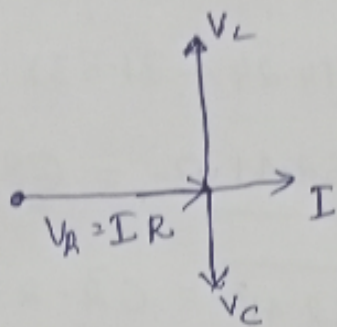
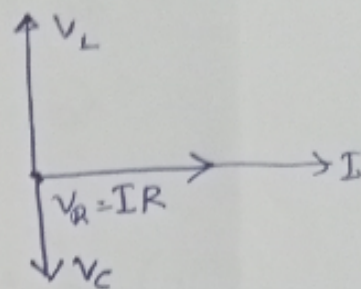


Let $V = V_m \sin \omega t$ be the alternating voltage

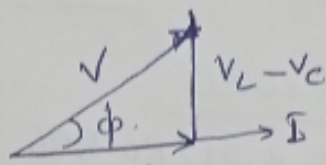
applied across the series RLC circuit. ③

Let V_R, V_L, V_C be the voltages across R, L and C respectively. Let I be the rms value of the resulting current.

$$V_R = IR, V_L = IX_L, V_C = IX_C.$$



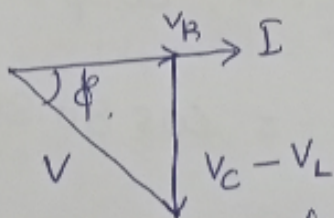
If $V_L > V_C$



Here I lags V by an angle ϕ .

$$\therefore i = I_m \sin(\omega t - \phi)$$

If $V_C > V_L$



Here I leads V by angle ϕ .

$$\therefore i = I_m \sin(\omega t + \phi)$$

So generally, $i = I_m \sin(\omega t \pm \phi)$

5. $R = 10 \Omega, L = 0.3 H, C = 100 \mu F = 100 \times 10^{-6} F$
 $V = 230 V, f = 50 Hz$.

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \text{--- (magnitude)} \quad (4)$$

$$Z = R + j(X_L - X_C) \text{ in complex form}$$

$$X_L = 2\pi fL = 2 \times 3.14 \times 50 \times 0.3 = \underline{94.24 \Omega}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 50 \times 100 \times 10^{-6}} = \underline{31.83 \Omega}$$

$$\begin{aligned} c) \quad Z &= 10 + j(94.24 - 31.83) \\ &= \underline{10 + j62.41 \Omega} = \underline{63.21 \angle 80.896^\circ} \end{aligned}$$

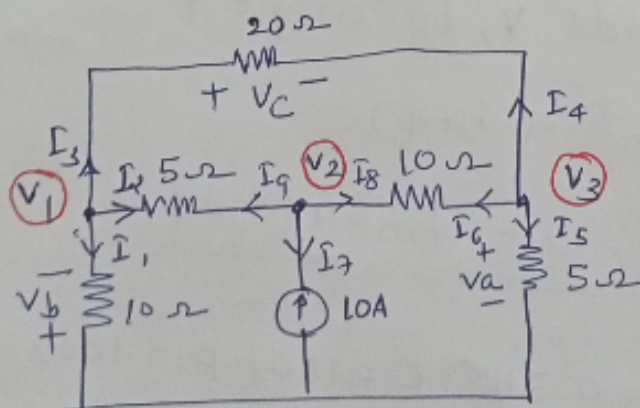
$$Z = \sqrt{10^2 + 62.41^2} = \underline{63.21 \Omega}$$

$$2) \quad i = \frac{V}{Z} = \frac{230 \angle 0}{63.21 \angle 80.896} = \underline{3.638 \angle -80.896^\circ A}$$

$$\text{magnitude of } i = \frac{230}{63.21} = \underline{3.638 A}$$

$$\begin{aligned} 3) \quad \text{power factor, } \cos \phi &= R/Z = \frac{10}{63.21} = \underline{0.1581 \text{ lag}} \\ &= \cos 80.896 = \underline{0.1581 \text{ lag}}. \end{aligned}$$

6.



(5)

At node 1

$$\hat{I}_1 + \hat{I}_2 + \hat{I}_3 = 0$$

$$\frac{V_1 - 0}{10} + \frac{V_1 - V_2}{5} + \frac{V_1 - V_3}{20} = 0$$

$$0.1V_1 + 0.2V_1 - 0.2V_2 + 0.05V_1 - 0.05V_3 = 0$$

$$0.35V_1 - 0.2V_2 - 0.05V_3 = 0 \quad \text{--- (1)}$$

At node 2

$$\hat{I}_7 + \hat{I}_8 + \hat{I}_9 = 0$$

$$\frac{V_2 - V_1}{5} + \frac{V_2 - V_3}{10} - 10 = 0$$

$$0.2V_2 - 0.2V_1 + 0.1V_2 - 0.1V_3 - 10 = 0 \quad \text{--- (2)}$$

$$-0.2V_1 + 0.3V_2 - 0.1V_3 = 10 \quad \text{--- (2)}$$

At node 3

$$\hat{I}_4 + \hat{I}_5 + \hat{I}_6 = 0$$

$$\frac{V_3 - V_1}{20} + \frac{V_3 - V_2}{10} + \frac{V_3 - 0}{5} = 0$$

$$0.05V_3 - 0.05V_1 + 0.1V_3 - 0.1V_2 + 0.2V_3 = 0$$

$$-0.05V_1 - 0.1V_2 + 0.35V_3 = 0 \quad \text{--- (3)}$$

Solving $V_1 = 45.45V$

$$V_2 = 72.72V$$

$$V_3 = 27.27V$$

$$V_a = V_3 = 27.27V$$

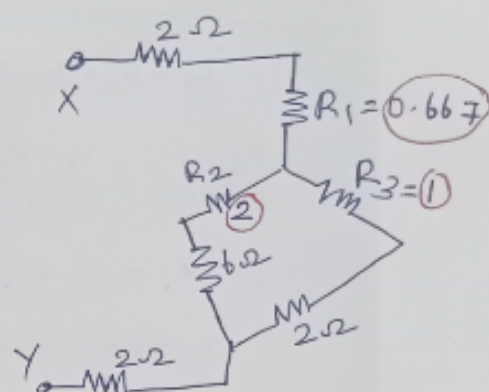
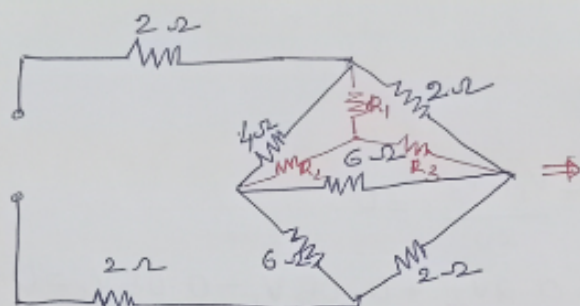
$$V_b = -V_1 = -45.45V$$

$$V_c = V_1 - V_3 = 18.18V$$

$$V_d = V_3 - V_2 = -45.46V$$

(6)

7.



$$R_1 = \frac{4 \times 2}{6 + 4 + 2} = \frac{8}{12} = 0.667 \Omega$$

$$R_2 = \frac{6 \times 4}{12} = \frac{24}{12} = 2 \Omega$$

$$R_3 = \frac{6 \times 2}{12} = \frac{12}{12} = 1 \Omega$$

R_2 and 6Ω in series = 8Ω

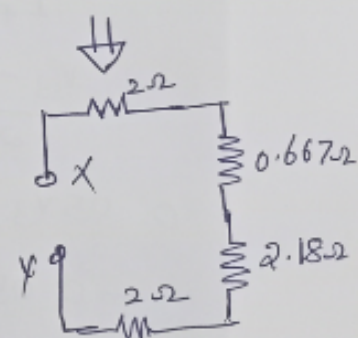
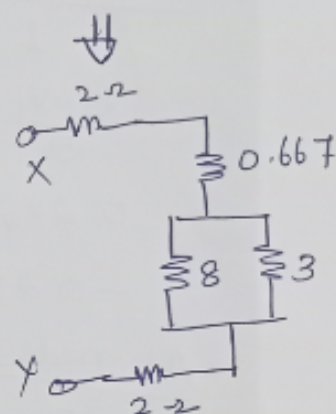
R_3 and 2Ω in series = 3Ω

8Ω and 3Ω in parallel =

$$= \frac{8 \times 3}{11} = 2.18 \Omega$$

$$\therefore R_{xy} = 2 + 0.667 + 2.18 + 2$$

$$= \underline{\underline{6.848 \Omega}}$$



8.

$$a = 3 \times 10^{-4} \text{ m}^2$$

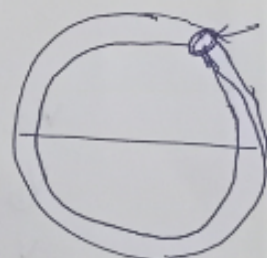
$$r = \frac{25}{2} \times 10^{-2} \text{ m} \quad d = 25 \times 10^{-2} \text{ m}$$

$$l_g = 0.5 \times 10^{-3} \text{ m}$$

$$N = 200$$

$$I = 3 \text{ A}$$

$$\phi = 0.28 \times 10^{-3} \text{ Wb}, \quad M_r = ?$$



$$l_i = \pi d - 0.5 \times 10^{-3} = \pi \times 25 \times 10^{-2} - 0.5 \times 10^{-3} = 0.7845 \text{ m}$$

$$l_a = 0.5 \times 10^{-3} \text{ m}$$

$$\text{mmf} = NI = 200 \times 3$$

Total mmf = mmf in the air gap + mmf in the iron path

$$NI = Hl = \frac{B}{\mu} \times l$$

$$200 \times 3 = H_a l_a + H_i l_i$$

$$= \frac{B}{\mu_0} l_a + \frac{B}{\mu_0 \mu_r} l_i$$

$$= \frac{B}{\mu_0} \left[l_a + \frac{l_i}{\mu_r} \right]$$

$$= \frac{0.933}{4\pi \times 10^{-7}} \left[0.5 \times 10^{-3} + \frac{0.7845}{\mu_r} \right]$$

$$\begin{aligned} B &= \mu H \\ B &= \frac{\phi}{A} \\ &= \frac{0.28 \times 10^{-3}}{3 \times 10^{-4}} \\ &= 0.933 \text{ T} \end{aligned}$$

$$\therefore \frac{200 \times 3 \times 4\pi \times 10^{-7}}{0.933} = 0.5 \times 10^{-3} + \frac{0.7845}{\mu_r}$$

$$\frac{0.7845}{\mu_r} = \frac{200 \times 3 \times 4\pi \times 10^{-7}}{0.933} - 0.5 \times 10^{-3} = 3.077 \times 10^{-4}$$

$$\therefore \mu_r = 2549.47$$

9. $v = 110 \sin 314t$

Comparing with standard value

$$v = V_m \sin \omega t$$

$$V_m = 110$$

$$\omega = 314$$

1) angular velocity, $\omega = 314 \text{ rad/second}$.

(8)

2) frequency = $\frac{\omega}{2\pi} = \frac{314}{2 \times 3.14} = 50 \text{ Hz}$

3) Time period = $\frac{1}{f} = \frac{1}{50} = 20 \text{ ms}$.

9.5) Statically induced emf & Dynamically Induced emf

The two types of induced emf are

i) Dynamically induced emf and

ii) statically induced emf. and statically induced emf are again classified into

a) self induced emf and

b) mutually induced emf.

Dynamically induced emf.

Here the field is stationary and conductor cut the magnetic field, an emf is induced in the conductor. Induced emf is given by,

$$e = Blv \sin \theta$$

If a conductor 'A' moves at an angle ' θ ' with the direction of flux (B tesla) with velocity v , and length of conductor l meters induced emf = $Blv \sin \theta$; and the direction of induced emf is given by Fleming's right hand rule.

Statically induced emf.

emf induced in a coil due to change in flux linkage, where the conductor is stationary.

a) Self induced emf

This is the emf induced in a coil due to the change of its own flux linked with it. If the current through the coil is changed, then the flux linked with its own turns will also change, which will produce in it what is called self induced emf. $e = L \frac{di}{dt}$

b) Mutually induced emf

An emf induced in one coil due to changing flux or current in the neighbouring coil is called mutually induced emf.

$$e = M \frac{dI_1}{dt}$$

10. $R = 10$; $L = 0.3H$, $C = 100 \times 10^{-6}$, $V = 230V$, $f = 50Hz$.

ref: Ques 5,

$$Z = 63.21 \angle 80.89$$

$$I = 3.369 \angle -80.89A$$

$$V_R = IR = 3.69 \times 10 = 36.39 \angle 80.89V$$

$$V_L = IX_L = 342.95V$$

$$V_C = IX_C = 115.856V$$

$$\text{Power consumed} = VI \cos \phi = 230 \times 3.369 \times \cos 80.89$$

$$2023/02/15 13:55 = 132.2W$$

$$10. \quad V_L = 400V$$

$$I_L = 20A$$

$$P_{3ph} = 10 \times 10^3$$

$$\sqrt{3} V_L I_L \cos \phi = 10 \times 10^3$$

$$\text{Power factor } \therefore \cos \phi = \frac{10 \times 10^3}{\sqrt{3} \times 400 \times 20} = \frac{10000}{13856.4} = \underline{\underline{0.721}}$$

$$\underline{\Delta} \quad V_L = 400V,$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}}$$

$$; \quad V_p = 400$$

$$I_p = \frac{20}{\sqrt{3}} = 11.547$$

$$Z_p = \frac{400}{11.547} = \underline{\underline{34.64 \Omega}}$$

Power consumed when the load is connected in Δ .

$$P = \sqrt{3} V_L I_L \cos \phi = 3 V_p I_p \cos \phi$$

$$\text{in Star connection, } V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.9V$$

$$I_p = \frac{V_{ph}}{Z} = \frac{230.9}{34.64} = 6.67A$$

$$\therefore \text{Total power consumed} = 3 V_p I_p \cos \phi = \underline{\underline{3335.05W}}$$